Experimenters never measure an infinite amount of anything. Dials never spin around to infinity. Meters never reach infinity. Calculators never register infinity. Almost always, an infinite answer is meaningless. All it tells us is that the equations of general relativity and quantum mechanics, when merged, go haywire.

Notice that this is quite unlike the tension between *special* relativity and quantum mechanics that came up in our discussion of quantum nonlocality in Chapter 4. There we found that reconciling the tenets of special relativity (in particular, the symmetry among all constant velocity observers) with the behavior of entangled particles requires a more complete understanding of the quantum measurement problem than has so far been attained (see this page–this page). But this incompletely resolved issue does not result in mathematical inconsistencies or in equations that yield nonsensical answers. To the contrary, the combined equations of special relativity and quantum mechanics have been used to make the most precisely confirmed predictions in the history of science. The quiet tension between special relativity and quantum mechanics points to an area in need of further theoretical development, but it has hardly any impact on their combined predictive power. Not so with the explosive union between general relativity and quantum mechanics, in which all predictive power is lost.

Nevertheless, you can still ask whether the incompatibility between general relativity and quantum mechanics really matters. Sure, the combined equations may result in nonsense, but when do you ever really need to use them together? Years of astronomical observations have shown that general relativity describes the macro world of stars, galaxies, and even the entire expanse of the cosmos with impressive accuracy; decades of experiments have confirmed that quantum mechanics does the same for the micro world of molecules, atoms, and subatomic particles. Since each theory works wonders in its own domain, why worry about combining them? Why not keep them separate? Why not use general relativity for things that are large and massive, quantum mechanics for things that are tiny and light, and celebrate humankind’s impressive achievement of successfully understanding such a wide range of physical phenomena?